

The Cardioid

1 Definition

A *cardioid* (from the Greek *kardia*, heart) is the plane curve traced by a point fixed on the circumference of a circle that rolls, without slipping, around a second circle of the same radius. It is therefore a special case of an *epicycloid*, the one with exactly one cusp.

2 Parametric equations

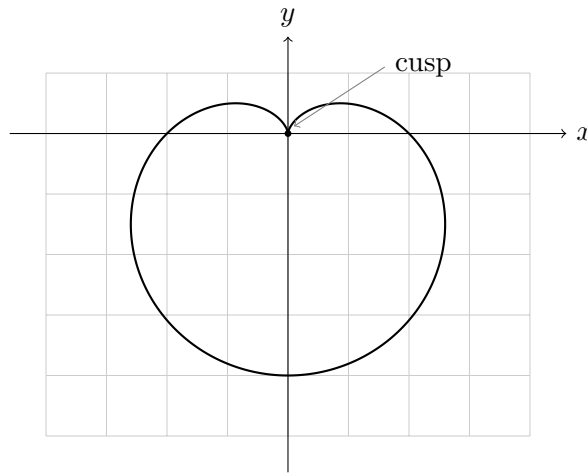
For a rolling radius a , the variant our program draws is

$$\begin{cases} x(\theta) = a(1 - \sin \theta) \cos \theta \\ y(\theta) = a(1 - \sin \theta) \sin \theta \end{cases} \quad \theta \in [0, 2\pi]. \quad (1)$$

Equivalently, in polar coordinates (r, θ) the curve is

$$r(\theta) = a(1 - \sin \theta), \quad (2)$$

which reads directly as: the distance from the origin swells to $2a$ at $\theta = -\frac{\pi}{2}$ (the bottom of the heart) and collapses to 0 at $\theta = \frac{\pi}{2}$, producing the cusp.



The cardioid $r = 1 - \sin \theta$ (that is, $a = 1$).

3 Properties

For the cardioid of parameter a :

- **Cusp.** At $\theta = \frac{\pi}{2}$ the factor $1 - \sin \theta$ vanishes, so the point reaches the origin with zero speed and reverses smoothly: the single sharp point characteristic of the curve.
- **Area.** The enclosed area is $\frac{1}{2} \int_0^{2\pi} r(\theta)^2 d\theta = \frac{3}{2}\pi a^2$ — exactly six times the area of the rolling circle of radius $a/2$ ¹.

¹With the rolling-circle construction, a cardioid of polar parameter a is generated by circles of radius $a/2$.

- **Arc length.** The total perimeter is $8a$, a pleasantly rational answer for such a round object.
- **Where you have seen one.** The bright curve at the bottom of a coffee cup lit from the side is (close to) a cardioid, and so is the pickup pattern of a “cardioid” microphone: sensitive in front, silent behind.

4 From mathematics to pixels

The OCaml program approximates the curve by a closed polyline: it samples $\theta_i = \frac{2\pi i}{n}$ for $i = 0, 1, \dots, n$, converts each $(x(\theta_i), y(\theta_i))$ to integer window coordinates, and joins consecutive points with `lineto`. For n around 200 the segments are shorter than a pixel almost everywhere, so the eye reads the polyline as a smooth curve.